

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z3,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z3 Z5: $\frac{Z_1 Z_3 Z_5 g_m - Z_1 Z_3}{2 Z_1 Z_3 g_m + Z_1 Z_5 g_m + Z_1 + Z_3 + Z_5}$

$$H(z) = \frac{Z_1 Z_3 Z_5 g_m - Z_1 Z_3}{2 Z_1 Z_3 g_m + Z_1 Z_5 g_m + Z_1 + Z_3 + Z_5}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m - 1}{C_1 C_3 R_5 s^2 + 2 g_m + s (C_1 + C_3 R_5 g_m + C_3)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_5} \sqrt{g_m}}{C_1 + C_3 R_5 g_m + C_3}$
 wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 + C_3 R_5 g_m + C_3}{C_1 C_3 R_5}$
 K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{C_1 C_3 R_3 R_5 s^2 + 2 R_3 g_m + R_5 g_m + s (C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_5} \sqrt{2 R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3}$
 wo: $\frac{\sqrt{2 R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3}{C_1 C_3 R_3 R_5}$
 K-LP: $\frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{C_1 C_3 R_1 R_5 s^2 + 2 R_1 g_m + s (C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_5} \sqrt{2 R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5}$
wo: $\frac{\sqrt{2 R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5}{C_1 C_3 R_1 R_5}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{C_1 C_3 R_1 R_3 R_5 s^2 + 2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5}$
wo: $\frac{\sqrt{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5}{C_1 C_3 R_1 R_3 R_5}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{C_3 C_5 R_1 R_5 s^2 + 2 R_1 g_m + s (C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{2 R_1 g_m + 1}}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2 R_1 g_m + 1}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5}}$
bandwidth: $\frac{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5}{C_3 C_5 R_1 R_5}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_5}{C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{C_3 C_5 R_1 R_3 s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_1g_m+1}}{C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1+C_5R_3} \\
\text{wo: } & \frac{\sqrt{R_1g_m+1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}} \\
\text{bandwidth: } & \frac{C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1+C_5R_3}{C_3C_5R_1R_3} \\
\text{K-LP: } & \frac{R_1R_3g_m}{R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_3}{C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1+C_5R_3} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3R_3s+1}, \infty, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_3R_5s + R_1R_3R_5g_m - R_1R_3}{C_3C_5R_1R_3R_5s^2 + 2R_1R_3g_m + R_1R_5g_m + R_1 + R_3 + R_5 + s(C_3R_1R_3R_5g_m + C_3R_1R_3 + C_3R_3R_5 + 2C_5R_1R_3R_5g_m + C_5R_1R_5 + C_5R_3R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_5}\sqrt{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}{C_3R_1R_3R_5g_m+C_3R_1R_3+C_3R_3R_5+2C_5R_1R_3R_5g_m+C_5R_1R_5+C_5R_3R_5} \\
\text{wo: } & \frac{\sqrt{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_1}\sqrt{R_3}\sqrt{R_5}} \\
\text{bandwidth: } & \frac{C_3R_1R_3R_5g_m+C_3R_1R_3+C_3R_3R_5+2C_5R_1R_3R_5g_m+C_5R_1R_5+C_5R_3R_5}{C_3C_5R_1R_3R_5} \\
\text{K-LP: } & \frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_3R_5}{C_3R_1R_3R_5g_m+C_3R_1R_3+C_3R_3R_5+2C_5R_1R_3R_5g_m+C_5R_1R_5+C_5R_3R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3R_3s+1}, \infty, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1R_3g_m + s(C_5R_1R_3R_5g_m - C_5R_1R_3)}{R_1g_m + s^2(C_3C_5R_1R_3R_5g_m + C_3C_5R_1R_3 + C_3C_5R_3R_5) + s(C_3R_1R_3g_m + C_3R_3 + 2C_5R_1R_3g_m + C_5R_1R_5g_m + C_5R_1 + C_5R_3 + C_5R_5) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_5}\sqrt{R_3}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}}{C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1R_5g_m+C_5R_1+C_5R_3+C_5R_5} \\
\text{wo: } & \sqrt{\frac{R_1g_m+1}{C_3C_5R_1R_3R_5g_m+C_3C_5R_1R_3+C_3C_5R_3R_5}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_3C_5R_1R_3R_5g_m+C_3C_5R_1R_3+C_3C_5R_3R_5}}(C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1R_5g_m+C_5R_1+C_5R_3+C_5R_5)}{\sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_3}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_3}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_3}\sqrt{C_5}\sqrt{R_3}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}} \\
\text{K-LP: } & \frac{R_1R_3g_m}{R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_5R_1R_3R_5g_m-C_5R_1R_3}{C_3R_1R_3g_m+C_3R_3+2C_5R_1R_3g_m+C_5R_1R_5g_m+C_5R_1+C_5R_3+C_5R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1s}, \infty, R_3, \infty, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_3s + R_3g_m}{C_1C_5R_3s^2 + g_m + s(C_1 + 2C_5R_3g_m + C_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_3}\sqrt{g_m}}{C_1+2C_5R_3g_m+C_5} \\
\text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_3}} \\
\text{bandwidth: } & \frac{C_1+2C_5R_3g_m+C_5}{C_1C_5R_3} \\
\text{K-LP: } & R_3 \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_3}{C_1+2C_5R_3g_m+C_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{C_1 C_5 R_3 R_5 s^2 + 2 R_3 g_m + R_5 g_m + s (C_1 R_3 + C_1 R_5 + 2 C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3} \sqrt{R_5} \sqrt{2 R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + 2 C_5 R_3 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2 R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_3 + C_1 R_5 + 2 C_5 R_3 R_5 g_m + C_5 R_5}{C_1 C_5 R_3 R_5}$
 K-LP: $\frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_3 R_5}{C_1 R_3 + C_1 R_5 + 2 C_5 R_3 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s (C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^2 (C_1 C_5 R_3 + C_1 C_5 R_5) + s (C_1 + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{g_m} \sqrt{R_3 + R_5}}{C_1 + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3 + C_1 C_5 R_5}}$
 bandwidth: $\frac{C_1 + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_3 + R_5} \sqrt{C_1 C_5 R_3 + C_1 C_5 R_5}}$
 K-LP: R_3
 K-HP: 0
 K-BP: $\frac{C_5 R_3 R_5 g_m - C_5 R_3}{C_1 + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5}$
 Qz: None
 Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2 g_m + s^2 (C_1 C_3 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5) + s (C_1 + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_5} \sqrt{g_m} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{C_1 + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
 wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1 C_3 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5}}$
 bandwidth: $\frac{C_1 + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}{\sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_5 + C_1 C_5 R_5 + C_3 C_5 R_5}}$
 K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5}{C_1 + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
 Qz: None
 Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{g_m + s^2 (C_1 C_3 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3) + s (C_1 + C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{g_m} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{C_1 + C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_3 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3}}$
 bandwidth: $\frac{C_1 + C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5}{\sqrt{R_3} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_3 + C_1 C_5 R_3 + C_3 C_5 R_3}}$

K-LP: R_3
K-HP: 0
K-BP: $-\frac{C_5 R_3}{C_1 + C_3 R_3 g_m + 2C_5 R_3 g_m + C_5}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s^2 (C_1 C_3 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5) + s (C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{2R_3 g_m + R_5 g_m + 1}}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2R_3 g_m + R_5 g_m + 1}}{\sqrt{C_1 C_3 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5}}$
bandwidth: $\frac{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}{\sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_3 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5}}$
K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_5}{C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_3 R_3 R_5 g_m - C_3 R_3) - 1}{2g_m + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5) + s (C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{g_m} \sqrt{R_3 + R_5}}{C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3}$
wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1 C_3 R_3 + C_1 C_3 R_5}}$
bandwidth: $\frac{C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_3 + R_5} \sqrt{C_1 C_3 R_3 + C_1 C_3 R_5}}$
K-LP: $\frac{R_5 g_m - 1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_3 R_3 R_5 g_m - C_3 R_3}{C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{C_1 C_5 R_1 R_3 s^2 + R_1 g_m + s (C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1}}{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3}}$
bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{C_1 C_5 R_1 R_3}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3}{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{C_1 C_5 R_1 R_3 R_5 s^2 + 2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s(C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{C_1 C_5 R_1 R_3 R_5}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_5}{C_1 R_1 R_3 + C_1 R_1 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + s(C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s^2(C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5) + s(C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3 + R_5} \sqrt{R_1 g_m + 1}}{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3 + R_5}}$
bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_3 + R_5} \sqrt{C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5}}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3}{C_1 R_1 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5}$
Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s^2(C_1 C_3 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5) + s(C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5}}$
bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{\sqrt{R_1} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_5 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_5}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_5}{C_1 R_1 + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{R_1 g_m + s^2(C_1 C_3 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3) + s(C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3}}$
bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}{\sqrt{R_1} \sqrt{R_3} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_3 + C_1 C_5 R_1 R_3 + C_3 C_5 R_1 R_3}}$

K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s^2 (C_1 C_3 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5) + s (C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}}{\sqrt{C_1 C_3 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5}}$
bandwidth: $\frac{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}{\sqrt{R_1} \sqrt{R_3} \sqrt{R_5} \sqrt{C_1 C_3 + C_1 C_5 + C_3 C_5} \sqrt{C_1 C_3 R_1 R_3 R_5 + C_1 C_5 R_1 R_3 R_5 + C_3 C_5 R_1 R_3 R_5}}$
K-LP: $\frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_3 R_5}{C_1 R_1 R_3 + C_1 R_1 R_5 + C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5 + 2C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3)}{2R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5) + s (C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 + R_5} \sqrt{2R_1 g_m + 1}}{C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5}$
wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5}}$
bandwidth: $\frac{C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3 + R_5} \sqrt{C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3}{C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5}$
Qz: None
Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2g_m + s^2 (C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5) + s (2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_5 g_m \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_5 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}}}{2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3}$
wo: $\sqrt{2} \sqrt{\frac{g_m}{C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5}} (2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3)}{\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_5 g_m \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_5 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}}}$
K-LP: $\frac{R_5 g_m - 1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_5 g_m - C_1 R_1}{2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3}$
Qz: None
Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3)}{2 R_3 g_m + R_5 g_m + s^2 (C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} R_5 g_m \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} R_5 \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3}$

wo: $\sqrt{\frac{2 R_3 g_m + R_5 g_m + 1}{C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5}}$

bandwidth: $\frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} R_5 g_m \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} R_5 \sqrt{\frac{2 R_3 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{\sqrt{\frac{2 R_3 g_m + R_5 g_m + 1}{C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5}} (2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3)}$

K-LP: $\frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + 1}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, R_3, \infty, R_5, \infty)$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, R_3, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 s + R_1 R_3 g_m}{R_1 g_m + s (2 C_5 R_1 R_3 g_m + C_5 R_1 + C_5 R_3) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, R_3, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_3 R_5 s + R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (2 C_5 R_1 R_3 R_5 g_m + C_5 R_1 R_5 + C_5 R_3 R_5)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, R_3, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + s (C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{R_1 g_m + s (2 C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + s (C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_5) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{C_3 C_5 R_1 s^2 + s (C_3 R_1 g_m + C_3 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2 R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (C_3 R_1 R_3 R_5 g_m + C_3 R_1 R_3 + C_3 R_3 R_5)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3)}{2 R_1 g_m + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5) + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m - C_5 R_1)}{s^2 (2 C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 + C_3 C_5 R_3) + s (C_3 R_1 g_m + C_3 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_3 C_5 R_1 R_3 R_5 g_m - C_3 C_5 R_1 R_3) + s (C_3 R_1 R_3 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (2 C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + C_3 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + s (C_1 R_3 + C_1 R_5) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2 (C_1 C_3 + C_1 C_5 + C_3 C_5) + s (C_3 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{C_1 C_3 C_5 R_5 s^3 + s^2 (C_1 C_3 + C_1 C_5 + C_3 C_5 R_5 g_m + C_3 C_5) + s (C_3 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_3 g_m + s (C_5 R_3 R_5 g_m - C_5 R_3)}{C_1 C_3 C_5 R_3 R_5 s^3 + g_m + s^2 (C_1 C_3 R_3 + C_1 C_5 R_3 + C_1 C_5 R_5 + C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3) + s (C_1 + C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 s^2 + g_m + s (C_3 R_3 g_m - C_5)}{C_1 C_3 C_5 R_3 s^3 + s^2 (C_1 C_3 + C_1 C_5 + 2 C_3 C_5 R_3 g_m + C_3 C_5) + s (C_3 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_5 s^2 + R_5 g_m + s (C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5) - 1}{C_1 C_3 C_5 R_3 R_5 s^3 + 2g_m + s^2 (C_1 C_3 R_3 + C_1 C_3 R_5 + C_1 C_5 R_5 + 2C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5) + s (C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3) + s (C_3 R_3 g_m + C_5 R_5 g_m - C_5)}{s^3 (C_1 C_3 C_5 R_3 + C_1 C_3 C_5 R_5) + s^2 (C_1 C_3 + C_1 C_5 + 2C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5) + s (C_3 g_m + 2C_5 g_m)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_5 g_m - R_1 R_3}{2R_1 R_3 g_m + R_1 R_5 g_m + R_1 + R_3 + R_5 + s (C_1 R_1 R_3 + C_1 R_1 R_5)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2 (C_1 C_3 R_1 + C_1 C_5 R_1 + C_3 C_5 R_1) + s (C_3 R_1 g_m + C_3 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_3 C_5 R_1 R_5 s^3 + s^2 (C_1 C_3 R_1 + C_1 C_5 R_1 + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m + s (C_5 R_1 R_3 R_5 g_m - C_5 R_1 R_3)}{C_1 C_3 C_5 R_1 R_3 R_5 s^3 + R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_5 R_1 R_3 + C_1 C_5 R_1 R_5 + C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_3 + C_3 C_5 R_3 R_5) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_5 R_1 R_3 g_m + C_5 R_1 R_5 g_m + C_5 R_1 + C_5 R_3 + C_5 R_5) + 1}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m - C_5 R_1)}{C_1 C_3 C_5 R_1 R_3 s^3 + s^2 (C_1 C_3 R_1 + C_1 C_5 R_1 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 + C_3 C_5 R_3) + s (C_3 R_1 g_m + C_3 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 - C_5 R_1 R_5)}{C_1 C_3 C_5 R_1 R_3 R_5 s^3 + 2R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_5 + C_3 C_5 R_3 R_5) + s (C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_3 C_5 R_1 R_3 R_5 g_m - C_3 C_5 R_1 R_3) + s (C_3 R_1 R_3 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3 (C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_1 R_5) + s^2 (C_1 C_3 R_1 + C_1 C_5 R_1 + 2C_3 C_5 R_1 R_3 g_m + C_3 C_5 R_1 R_5 g_m + C_3 C_5 R_1 + C_3 C_5 R_3 + C_3 C_5 R_5) + s (C_3 R_1 g_m + C_3 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3)}{2R_3 g_m + R_5 g_m + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5) + 1}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s(C_1 R_1 g_m - C_5)}{C_1 C_3 C_5 R_1 s^3 + s^2(C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_5 R_1 g_m + C_1 C_5 + C_3 C_5) + s(C_3 g_m + 2C_5 g_m)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s(C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{C_1 C_3 C_5 R_1 R_5 s^3 + 2g_m + s^2(C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + C_3 C_5 R_5) + s(2C_1 R_1 g_m + C_1 + C_3 R_5 g_m + C_3 + 2C_5 R_5 g_m)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^2(C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s(C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^3(C_1 C_3 C_5 R_1 R_5 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_5) + s^2(C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_5 R_1 g_m + C_1 C_5 + C_3 C_5 R_5 g_m + C_3 C_5) + s(C_3 g_m + 2C_5 g_m)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 s^2 + R_3 g_m + s(C_1 R_1 R_3 g_m - C_5 R_3)}{C_1 C_3 C_5 R_1 R_3 s^3 + g_m + s^2(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_3 C_5 R_3) + s(C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_5 R_3 g_m + C_5)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_5 s^2 + R_3 R_5 g_m - R_3 + s(C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 - C_5 R_3 R_5)}{C_1 C_3 C_5 R_1 R_3 R_5 s^3 + 2R_3 g_m + R_5 g_m + s^2(C_1 C_3 R_1 R_3 R_5 g_m + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 R_5 + 2C_1 C_5 R_1 R_3 R_5 g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_3 R_5 + C_3 C_5 R_3 R_5) + s(2C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + C_3 R_3 R_5 g_m + C_3 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s^2(C_1 C_5 R_1 R_3 R_5 g_m - C_1 C_5 R_1 R_3) + s(C_1 R_1 R_3 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3(C_1 C_3 C_5 R_1 R_3 R_5 g_m + C_1 C_3 C_5 R_1 R_3 + C_1 C_3 C_5 R_3 R_5) + s^2(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_1 C_5 R_5 + C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3) + s(C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 s^3 + g_m + s^2(C_1 C_3 R_1 R_3 g_m - C_1 C_5 R_1 - C_3 C_5 R_3) + s(C_1 R_1 g_m + C_3 R_3 g_m - C_5)}{s^3(2C_1 C_3 C_5 R_1 R_3 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_3) + s^2(C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_3 C_5 R_3 g_m + C_3 C_5) + s(C_3 g_m + 2C_5 g_m)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_3 C_5 R_1 R_3 R_5 s^3 + R_5 g_m + s^2(C_1 C_3 R_1 R_3 R_5 g_m - C_1 C_3 R_1 R_3 - C_1 C_5 R_1 R_5 - C_3 C_5 R_3 R_5) + s(C_1 R_1 R_5 g_m - C_1 R_1 + C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5) - 1}{2g_m + s^3(2C_1 C_3 C_5 R_1 R_3 R_5 g_m + C_1 C_3 C_5 R_1 R_5 + C_1 C_3 C_5 R_3 R_5) + s^2(2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5) + s(2C_1 R_1 g_m + C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_5 R_5 g_m)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^3(C_1 C_3 C_5 R_1 R_3 R_5 g_m - C_1 C_3 C_5 R_1 R_3) + s^2(C_1 C_3 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3) + s(C_1 R_1 g_m + C_3 R_3 g_m + C_5 R_5 g_m - C_5)}{s^3(2C_1 C_3 C_5 R_1 R_3 g_m + C_1 C_3 C_5 R_1 R_5 g_m + C_1 C_3 C_5 R_1 + C_1 C_3 C_5 R_3 + C_1 C_3 C_5 R_5) + s^2(C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5) + s(C_3 g_m + 2C_5 g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_3 C_5 R_1 R_3 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_3 R_1 R_3 R_5 g_m - C_3 R_1 R_3 - C_5 R_1 R_5)}{2 R_1 g_m + s^2 (2 C_3 C_5 R_1 R_3 R_5 g_m + C_3 C_5 R_1 R_5 + C_3 C_5 R_3 R_5) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_5 g_m + C_3 R_1 + C_3 R_3 + C_3 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_3}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}}{2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_1g_m+1}{2C_3C_5R_1R_3R_5g_m+C_3C_5R_1R_5+C_3C_5R_3R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+1}{2C_3C_5R_1R_3R_5g_m+C_3C_5R_1R_5+C_3C_5R_3R_5}}(2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_5R_1R_5g_m+C_5R_5)}{2\sqrt{C_3}\sqrt{C_5}R_1R_3\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_3}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_3}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}} \\ \text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\ \text{K-HP: } & -\frac{R_1R_3}{2R_1R_3g_m+R_1+R_3} \\ \text{K-BP: } & \frac{C_3R_1R_3R_5g_m-C_3R_1R_3-C_5R_1R_5}{2C_3R_1R_3g_m+C_3R_1R_5g_m+C_3R_1+C_3R_3+C_3R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}} \end{aligned}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m - C_5 R_3)}{g_m + s^2 (2 C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 + C_1 C_5 R_3) + s (C_1 R_1 g_m + C_1 + 2 C_5 R_3 g_m + C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}}{C_1R_1g_m+C_1+2C_5R_3g_m+C_5} \\ \text{wo: } & \sqrt{\frac{g_m}{2C_1C_5R_1R_3g_m+C_1C_5R_1+C_1C_5R_3}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{2C_1C_5R_1R_3g_m+C_1C_5R_1+C_1C_5R_3}}(C_1R_1g_m+C_1+2C_5R_3g_m+C_5)}{2\sqrt{C_1}\sqrt{C_5}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1+R_3}}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & -\frac{R_1R_3}{2R_1R_3g_m+R_1+R_3} \\ \text{K-BP: } & \frac{C_1R_1R_3g_m-C_5R_3}{C_1R_1g_m+C_1+2C_5R_3g_m+C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}} \end{aligned}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_3 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_1 R_1 R_3 R_5 g_m - C_1 R_1 R_3 - C_5 R_3 R_5)}{2 R_3 g_m + R_5 g_m + s^2 (2 C_1 C_5 R_1 R_3 R_5 g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_3 R_5) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_5 g_m + C_1 R_1 + C_1 R_3 + C_1 R_5 + 2 C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{R_5}g_m\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}}{2C_1R_1R_3g_m+C_1R_1R_5g_m+C_1R_1+C_1R_3+C_1R_5+2C_5R_3R_5g_m+C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_1C_5R_1R_3R_5g_m+C_1C_5R_1R_5+C_1C_5R_3R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_1C_5R_1R_3R_5g_m+C_1C_5R_1R_5+C_1C_5R_3R_5}}(2C_1R_1R_3g_m+C_1R_1R_5g_m+C_1R_1+C_1R_3+C_1R_5+2C_5R_3R_5g_m+C_5R_5)}{2\sqrt{C_1}\sqrt{C_5}R_1R_3\sqrt{R_5}g_m\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_1R_3g_m+R_1+R_3}+\frac{R_5g_m}{2R_1R_3g_m+R_1+R_3}+\frac{1}{2R_1R_3g_m+R_1+R_3}}} \\ \text{K-LP: } & \frac{R_3R_5g_m-R_3}{2R_3g_m+R_5g_m+1} \\ \text{K-HP: } & -\frac{R_1R_3}{2R_1R_3g_m+R_1+R_3} \\ \text{K-BP: } & \frac{C_1R_1R_3R_5g_m-C_1R_1R_3-C_5R_3R_5}{2C_1R_1R_3g_m+C_1R_1R_5g_m+C_1R_1+C_1R_3+C_1R_5+2C_5R_3R_5g_m+C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s^2 (C_1 C_5 R_1 R_3 R_5 g_m - C_1 C_5 R_1 R_3) + s (C_1 R_1 R_3 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^2 (2C_1 C_5 R_1 R_3 g_m + C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_1 + C_1 C_5 R_3 + C_1 C_5 R_5) + s (C_1 R_1 g_m + C_1 + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}}{C_1R_1g_m+C_1+2C_5R_3g_m+C_5R_5g_m+C_5}$$

$$\text{wo: } \sqrt{\frac{g_m}{2C_1C_5R_1R_3g_m+C_1C_5R_1R_5g_m+C_1C_5R_1+C_1C_5R_3+C_1C_5R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{g_m}{2C_1C_5R_1R_3g_m+C_1C_5R_1R_5g_m+C_1C_5R_1+C_1C_5R_3+C_1C_5R_5}}(C_1R_1g_m+C_1+2C_5R_3g_m+C_5R_5g_m+C_5)}{2\sqrt{C_1}\sqrt{C_5}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}}$$

K-LP: R_3

$$\text{K-HP: } \frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}$$

$$\text{K-BP: } \frac{C_1R_1R_3g_m+C_5R_3R_5g_m-C_5R_3}{C_1R_1g_m+C_1+2C_5R_3g_m+C_5R_5g_m+C_5}$$

Qz: None

$$\text{Wz: } \sqrt{\frac{g_m}{C_1C_5R_1R_5g_m-C_1C_5R_1}}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_3 R_1 R_3 R_5 g_m - C_1 C_3 R_1 R_3) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_3 R_3 R_5 g_m - C_3 R_3) - 1}{2g_m + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_5 g_m + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_3 R_3 g_m + C_3 R_5 g_m + C_3)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_5\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}}{2C_1R_1g_m+C_1+2C_3R_3g_m+C_3R_5g_m+C_3}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_1C_3R_1R_3g_m+C_1C_3R_1R_5g_m+C_1C_3R_1+C_1C_3R_3+C_1C_3R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_3R_1R_3g_m+C_1C_3R_1R_5g_m+C_1C_3R_1+C_1C_3R_3+C_1C_3R_5}}(2C_1R_1g_m+C_1+2C_3R_3g_m+C_3R_5g_m+C_3)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_3g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_5\sqrt{\frac{g_m}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}}}$$

$$\text{K-LP: } \frac{R_5g_m-1}{2g_m}$$

$$\text{K-HP: } \frac{R_1R_3R_5g_m-R_1R_3}{2R_1R_3g_m+R_1R_5g_m+R_1+R_3+R_5}$$

$$\text{K-BP: } \frac{C_1R_1R_5g_m-C_1R_1+C_3R_3R_5g_m-C_3R_3}{2C_1R_1g_m+C_1+2C_3R_3g_m+C_3R_5g_m+C_3}$$

Qz: None

$$\text{Wz: } \frac{1}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}}$$

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